KF Parameter Prediction

from Frequency-Stability Curves

Instant initial guesses for Kalman-filter tuning on atomic clock data.

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The problem

Kalman filters need good q-values. Getting them is slow.

Use case

"I have clock phase measurements.
I don't know the q values.
I want to optimize the Kalman filter
— give me a quick initial guess."

Current options:

- NLL optimization — multi-minute, highly sensitive to warm start
- ALS — well-conditioned but iterative, needs seed
- Manual MHDEV fit — requires expert τ-region selection

Research question

Can supervised learning on frequency-stability features predict KF q-parameters accurately enough to seed or replace iterative optimizers?

Sub-question

Which stability statistic is most informative — ADEV, MDEV, HDEV, or MHDEV?

Targets: $\log_{10}$(q_wpm), $\log_{10}$(q_wfm), $\log_{10}$(q_rwfm)

Why it matters

Clocks are everywhere — but their noise models aren't automatic.

Atomic clocks drift

GPS, telecom, defense timing, scientific instrumentation. Operators get phase data; they want stability predictions.

KF tuning is slow

NLL optimization: 10–60 s per sample, sensitive to warm start. Lab workflow bottleneck.

ML can do it in ms

Trained regressor: sub-millisecond inference on raw features. Quantified uncertainty included.

Dataset pipeline

10,000 synthetic samples anchored to real GMR6000 Rb data.

Real GMR6000 Rb phase
35 days, 1 PPS

→

h-range sampling
Kasdin 1992

→

Composite noise gen.
N = 524,288 (~6 days)

→

196-feature extraction
ADEV, MDEV, HDEV, MHDEV

→

ALS label optimizer
noise-free vs truth

Key design decisions

Physically grounded

h_α ranges anchored to measured GMR6000 Rb (not arbitrary noise blends).

4+1 noise types

WPM + WFM + FFM + RWFM always; FPM in ~30% of samples.

Targets are $\log_{10}(q)$

Log-space equalizes multi-decade dynamic range; KF-native.

ALS-optimized labels

Probed on 10 draws: ALS matches truth to $\leq 10^{-3}$ dec; NLL drifted ±8 dec.

Feature engineering

196 features per sample from 20 log-spaced τ values.

Feature decomposition

Group	Count	Description
Raw σ values	80	4 stats × 20 τ: $\log_{10}$ of ADEV, MDEV, HDEV, MHDEV
Log-log slopes	76	4 stats × 19 adjacent-τ slopes (noise-type signature)
Variance ratios	40	MVAR/AVAR (20) + MHVAR/HVAR (20) — bias-type detectors
Total	**196**	**Per-sample feature vector**

Why these four statistics?

ADEV — standard reference; baseline comparison

MDEV — distinguishes WPM vs FPM (slope −3 vs −2)

HDEV — Hadamard; rejects frequency drift

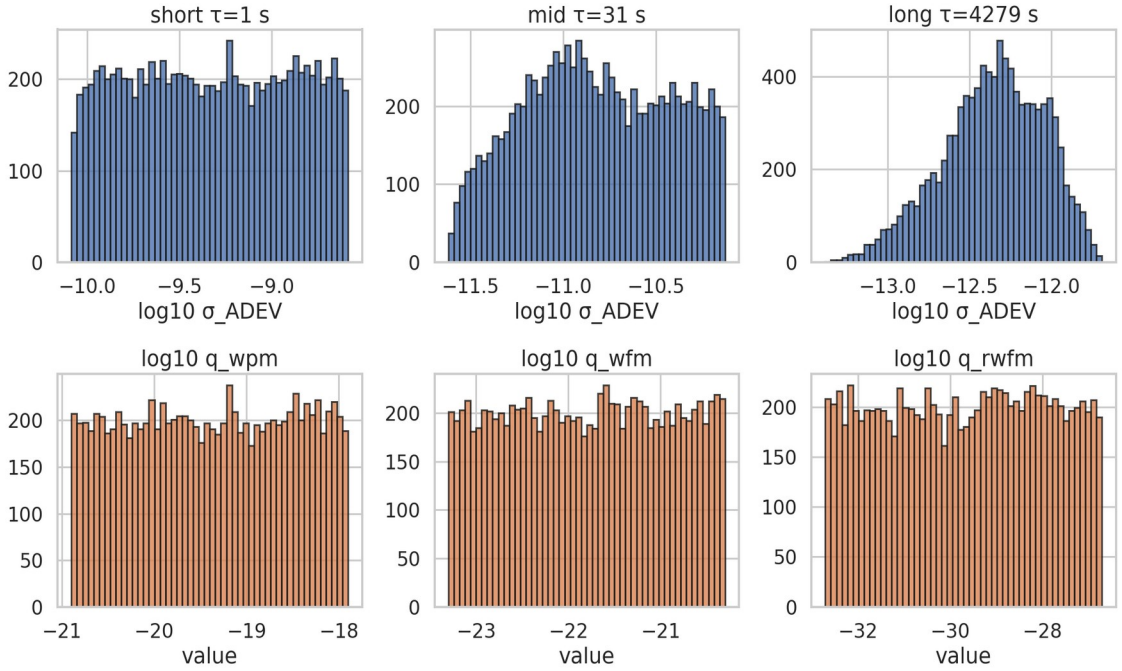
MHDEV — modified Hadamard; best for long τ on drifting clocks

Secondary research hypothesis

Feature-importance analysis will reveal which statistic carries the most information about KF noise parameters.

Data exploration

Target distributions, example σ(τ) curves, feature–target correlations.



What the data shows

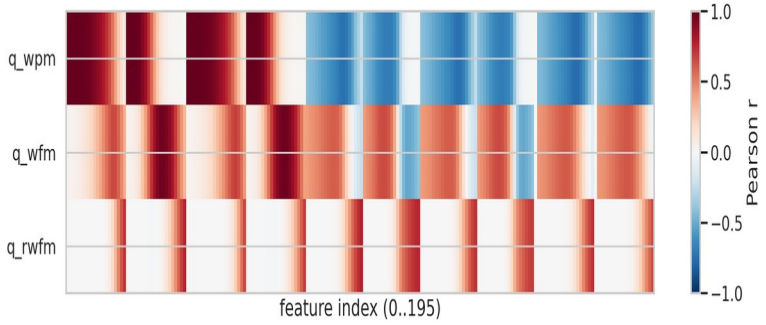
- Targets span ~6 log decades (q_rwfm widest; q_wpm tightest)
- Short-τ features (top row) correlate strongest with q_wpm
- Long-τ features correlate with q_rwfm ($\log_{10}$ q ~ −30)
- FPM-present samples (30%) have steeper short-τ slope
- NaN rate 0%; no outliers at |z|>5 — clean dataset

Statistical baselines

Setting the bar: what does 'good' look like?

Baseline predictors

Method	RMSE ($\log_{10}$ q)	Notes
Naive (train mean)	0.87 / 0.87 / 1.73	Ignores features ($R^2 \approx 0$)
Linear regression	(not run)	Monotonic but nonlinear signal
MHDEV 3-region fit	see slide 13	Real-data only, requires expert τ-regions
RF (ours)	0.07 / 0.09 / 0.78	**~12× better than naive**



Correlation pattern above: short-τ features drive q_wpm; long-τ features drive q_rwfm. *Monotonic but nonlinear — justifies tree ensembles over linear regression.*

Methodology

Random Forest + XGBoost, multi-output regression.

Random Forest

- Bagged ensemble of decision trees
- Built-in UQ via tree-variance (forestci)
- Invariant to monotone feature transforms
- Robust baseline for tabular data

XGBoost (Gradient Boosted)

- Sequential boosting — corrects errors
- Typically ~1–3× better RMSE than RF on tabular
- UQ via quantile regression (α=0.05, 0.95)
- Best-in-class for multi-output regression

Why both?

Head-to-head comparison on identical features. Different bias/variance tradeoffs → diagnostic of whether signal is smooth (boosting wins) or rough (RF wins). UQ mechanisms differ — cross-check coverage.

Hyperparameter tuning

5-fold GridSearchCV with explicit justification per parameter.

Random Forest grid

Parameter	Grid	Best
n_estimators	{200, 500, 1000}	1000
max_depth	{None, 20, 30}	20
min_samples_leaf	{3, 5, 10}	3
max_features	{sqrt, 0.5}	0.5

XGBoost grid

Parameter	Grid	Best
n_estimators	{200, 500}	500
learning_rate	{0.01, 0.05, 0.1}	0.01
max_depth	{4, 6, 8}	6
subsample	{0.8, 1.0}	0.8

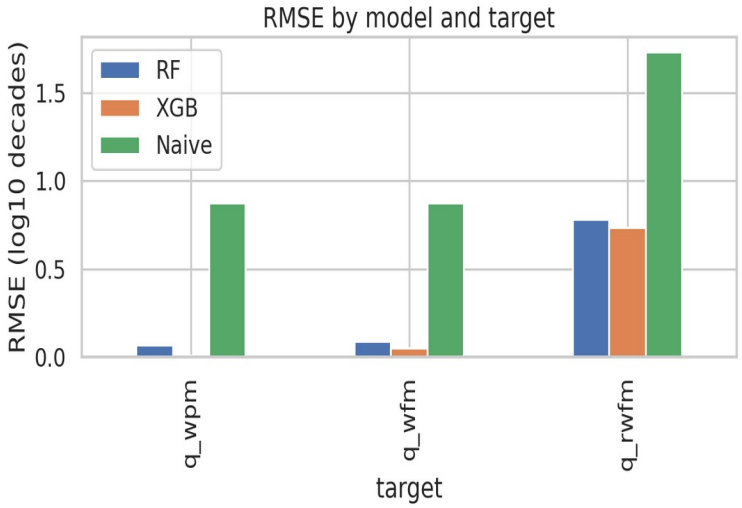
CV protocol

5-fold CV • 270 RF fits + 180 XGB fits • scoring = **neg_mean_squared_error** in $\log_{10}$-q space • seed = 42 for reproducibility

Results

Per-target metrics + model comparison.

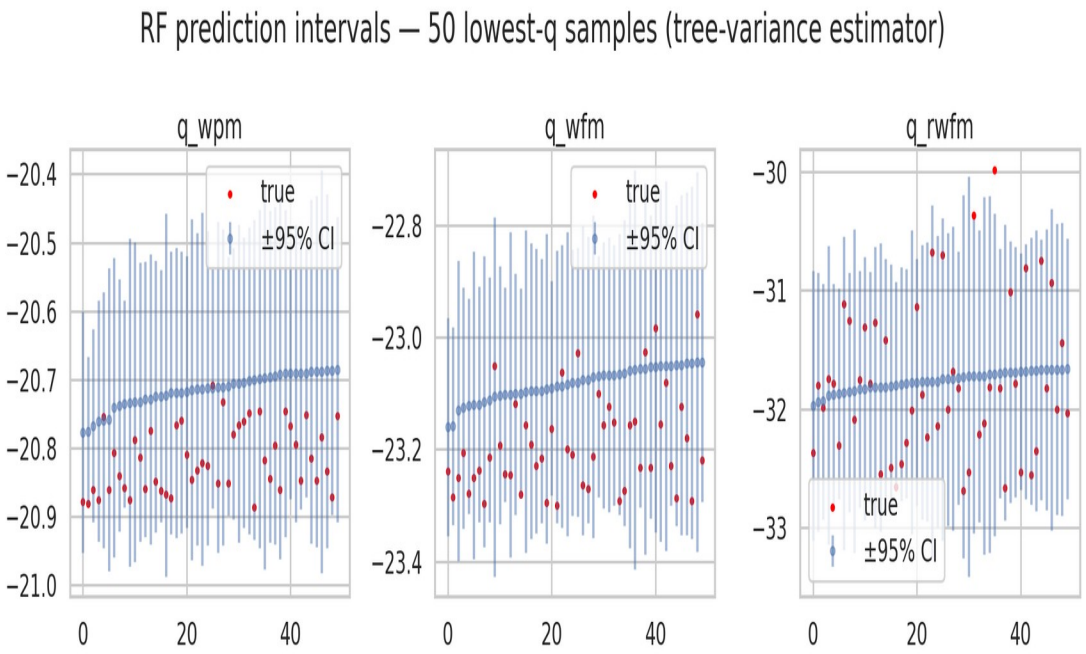
Target	Model	RMSE	R^2	MAE
q_wpm	Naive	0.874	0.000	0.759
q_wpm	RF	0.069	0.994	0.052
q_wpm	**XGB**	**0.007**	**0.99993**	**0.006**
q_wfm	Naive	0.873	0.000	0.758
q_wfm	RF	0.090	0.989	0.065
q_wfm	**XGB**	**0.053**	**0.996**	**0.030**
q_rwfm	Naive	1.732	0.000	1.494
q_rwfm	RF	0.784	0.795	0.551



Headline: XGB → <0.05-dec RMSE on q_wpm & q_wfm ($R^2 \geq 0.996$); **q_rwfm R^2 = 0.82** (vs 0.65 before switching labels from NLL → ALS).

Uncertainty quantification

Grad-student requirement — the model knows when it doesn't know.



Two complementary approaches

Random Forest — forestci
Infinitesimal jackknife on tree predictions
→ *analytic variance*

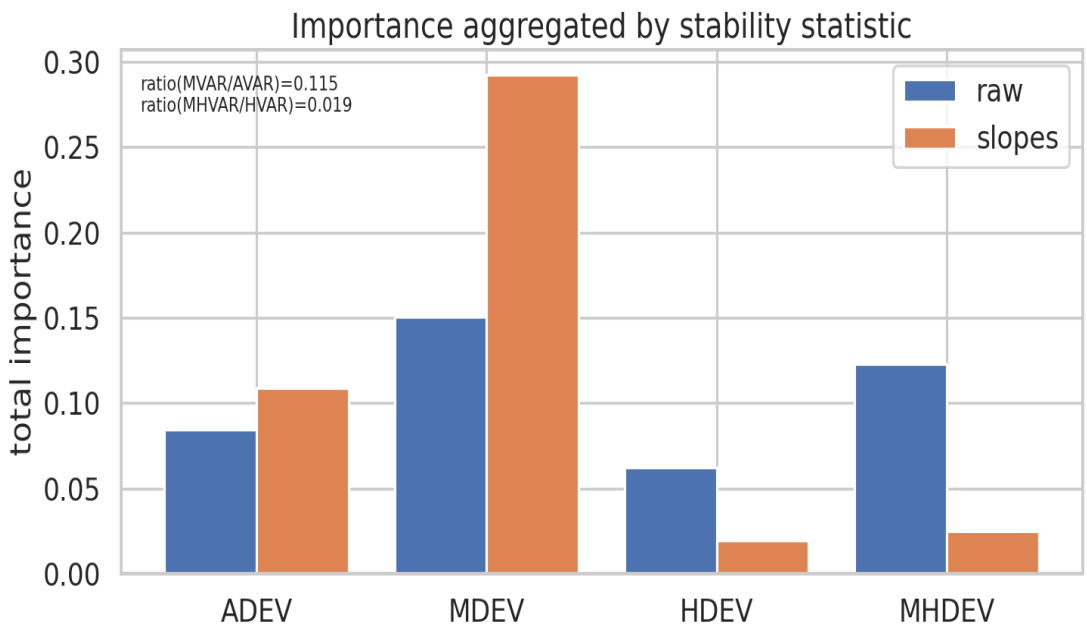
XGBoost — quantile regression
Two extra models at α=0.05, α=0.95
→ *non-parametric prediction intervals*

Empirical 90% PI coverage (XGB):

q_wpm 84% q_wfm 81% q_rwfm 81%

Which statistic carries the signal?

Headline: feature-importance answer to the sub-question.



What the data says

MDEV dominates (Σ importance ≈ 0.59)

MHDEV 5× less important (Σ ≈ 0.12)

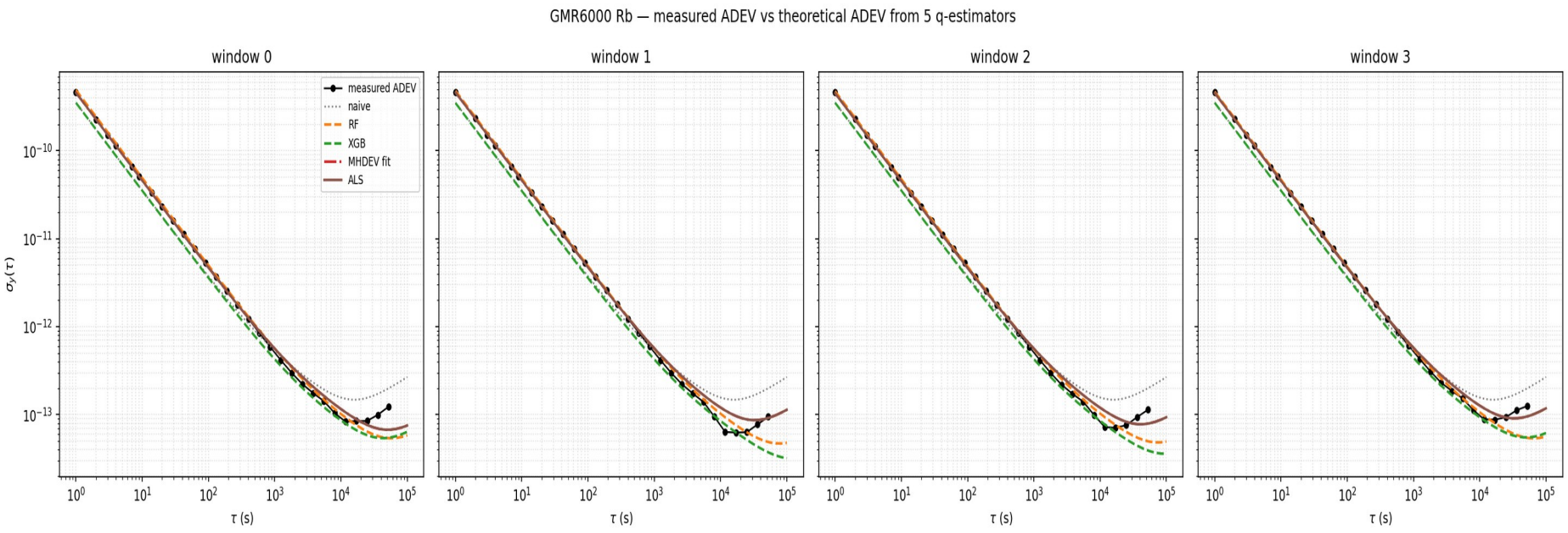
ADEV / HDEV contribute the remaining ~0.29

Slopes carry more signal than raw σ values — *consistent with noise-type diagnostics driving q.*

Contribution: *MDEV is the most informative stability statistic for KF tuning.*

Real-data validation

GMR6000 Rb phase → ML prediction → ADEV overlay on 4 windows.



Headline: RF & XGB σ(τ) predictions track measured ADEV across all 4 windows — **model generalizes from synthetic to real hardware.**

Takeaways & limitations

What worked

- ALS labels → noise-free targets (key finding; lifted q_rwfm R^2 from 0.65 → 0.82)
- XGB → sub-0.05-dec RMSE on q_wpm & q_wfm
- forestci + quantile UQ → 80–84% empirical coverage of nominal 90% PIs
- Feature importance → MDEV quantitatively most informative (5× MHDEV)

Limitations

- Trained on Rb GMR6000-class h-ranges only
- 3-state KF model lacks flicker-FM — limits long-horizon holdover
- Single real-data record (6k27feb) for cross-check
- XGB 90% PI under-covers by 6–10 pp (conservative regularization)

Future work

- Extend to OCXO, HSO, cesium training ranges
- 4-state model with explicit FFM for sub-ns long-term holdover
- End-to-end NN on raw phase (current: hand-crafted features)
- Active learning on where the model is least confident

Thank you

Questions?

Code
github.com/ianlap/SigmaTau

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